

1. Introduction. This program prints the primes below *limit*, using the sieve of Eratosthenes. We state the overall shape of the program first and defer every detail to a later section.

```
define limit = 50
program primes(output);
  var ⟨ Global variables 2 ⟩
  begin ⟨ Cross out the composites 3 ⟩;
    ⟨ Print the survivors 4 ⟩;
  end.
```

2. The sieve. The array *composite* records what we have crossed out.

```

⟨Global variables 2⟩ ≡
composite: array [2 .. limit] of boolean;
p, q, n: integer;

```

This code is used in section 1.

3. We need only cross out multiples of p starting at p^2 , because every smaller multiple has a smaller factor and is already gone.

```

⟨Cross out the composites 3⟩ ≡
  for  $n \leftarrow 2$  to limit do composite[ $n$ ]  $\leftarrow$  false;
   $p \leftarrow 2$ ;
  while  $p * p \leq$  limit do
    begin if  $\neg$ composite[ $p$ ] then
      begin  $q \leftarrow p * p$ ;
      while  $q \leq$  limit do
        begin composite[ $q$ ]  $\leftarrow$  true;  $q \leftarrow q + p$ ;
        end;
      end;
     $p \leftarrow p + 1$ ;
  end

```

This code is used in section 1.

4. Whatever survives is prime.

```

⟨Print the survivors 4⟩ ≡
  for  $n \leftarrow 2$  to limit do
    if  $\neg$ composite[ $n$ ] then write( $n : 4$ );
  writeln

```

This code is used in section 1.

```

boolean: 2.
composite: 2, 3, 4.
false: 3.
integer: 2.
limit: 1, 2, 3, 4.
output: 1.
primes: 1.
true: 3.
write: 4.
writeln: 4.

```

〈Cross out the composites 3〉 Used in section 1.
 〈Global variables 2〉 Used in section 1.
 〈Print the survivors 4〉 Used in section 1.

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